

OSBORNE V. LOPES

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EDUCATION

Northeastern University, USA

Dec 2025

Master of Science, Information Systems

Coursework: Big Data and Intelligent Analytics, Data Architecture and Business Intelligence, Data Science Methods and Tools

University of Mumbai, India

Jun 2021

Bachelor of Technology, Information Technology

Coursework: Data Mining and Business Intelligence, Database Management, Cloud Computing Services

EXPERIENCE

Saayam For All

Feb 2026 - Present

Data Engineer

San Jose, CA

- Built scalable data pipelines in **Microsoft Fabric Lakehouse** to ingest and transform volunteer, service request and donor data across 12+ language channels, reducing data processing time by 45% and enabling real-time operational dashboards
- Automated ETL orchestration and data quality workflows using **Control-M** and **Alteryx**, reducing manual data reconciliation by 15 hrs/week and improving accuracy of beneficiary and impact reporting

Atlas SP

Jan 2025 - Aug 2025

Data Engineer Co-op

New York, NY

- Implemented **ADF** pipelines to automate ingestion of millions of records from multiple sources, reducing processing time by 45%
- Stored processed data in Azure Blob Storage and Azure AI Search vector indexes, enabling a **RAG-based chatbot** served via **FastAPI** for internal teams to query financial data, reducing repetitive analyst requests by 30%
- Designed **normalized and dimensional data models** in Snowflake and optimized Delta Lake queries to retrieve data only on market days, improving query performance by 20% and enabling cleaner portfolio analytics for executive reporting
- Optimized streaming workflows using Kafka and **Databricks**, enabling sub-minute data refresh and 3x scalability during peak trading hours

Ingram Micro

Oct 2022 - Aug 2023

Analytics Engineer

Mumbai, India

- Designed **star-schema semantic models** in Power BI with advanced DAX measures to track inventory turnover, order fulfillment and supplier performance, earning trust of cross-functional logistics leadership through executive-level KPI reporting across 5+ business teams
- Reduced inventory shortfalls by 20% by owning end-to-end **demand forecasting pipelines** integrating historical shipment data, supplier lead times and seasonal trends into supply chain planning
- Analyzed customer order and delivery performance metrics** across 10+ enterprise clients via incremental DBT models and Looker dashboards, conducting **A/B tests** on reporting formats to identify SLA breach trends and reducing repeat complaints by 15%

ICICI Securities

Jun 2021 - Sep 2022

Data Analyst

Mumbai, India

- Partnered with trading and operations stakeholders to deliver structured **Tableau** dashboards using Python and **Redis** caching, cutting report processing time by 50% and saving 8 hours weekly across trade and portfolio analytics
- Dived deep into trading datasets using Teradata to build reusable SQL models using **window functions and partition pruning**, reducing processing time by 40% and owning end-to-end dataset reliability for 5+ business teams

PROJECTS

RAG Job Assistant | [GitHub](#)

May 2025

- Built a production-grade **RAG pipeline** indexing 5,000+ job postings into **ChromaDB** using **HuggingFace** sentence-transformer embeddings (all-MiniLM-L6-v2), enabling **semantic search** over job descriptions
- Designed an end-to-end ingestion pipeline (SQLite to HTML stripping to **vector embeddings** to ChromaDB) with incremental updates; integrated **LangChain** and Groq API (Llama 3.3 70B) for LLM-powered answer generation

NewsScope | [GitHub](#)

Dec 2024

- Built NewsScope, an industry-specific **news aggregator** that scrapes, processes and surfaces real-time business news using Python (BeautifulSoup, Scrapy) and **DBT** on **BigQuery**
- Designed an end-to-end ETL pipeline orchestrated via **Airflow**, landing raw data in **GCS** and transforming in **BigQuery**, with schema validation using Pydantic 2 and data quality checks via **Great Expectations**
- Containerized and deployed the platform with Docker, exposing a Streamlit interface for browsing curated news by industry

SKILLS

Languages: Python, Java, R, SQL, Pro*C

Databases and Warehousing: Snowflake, Teradata, MongoDB, DynamoDB, Redshift

Cloud Platforms: Azure (ADF, Blob Storage, Synapse, Fabric), AWS (S3, Glue, EMR, Lambda, Athena, Kinesis, IAM), GCP (BigQuery, DataFlow, GCS), Docker

Tools: Airflow, DBT, Apache Spark, Kafka, Databricks, Alteryx, Control-M, Git

Analytical Toolkit: Tableau, PowerBI, Looker, Google Data Studio, QuickSight, Excel

AI and Automation: RAG, Agentic AI, LangChain, HuggingFace, FastAPI, PineCone, LLM, n8n, Prompt Engineering